

Varenya Sri Mudumba

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Education

Texas A&M University, College Station, TX

Aug 2025 - May 2027

M.S. in Computer Science - GPA 4.0/4.0 (Expected May 2027)

National Institute of Technology, Durgapur

Dec 2020 - May 2024

B.Tech in Computer Science and Engineering - GPA 8.84/10.00

Technical Skills

Languages: C/C++, Python, Go, Rust, JavaScript, SQL, Bash, Groovy.

Backend & Web: REST APIs, gRPC, Node.js, Express, React.js, microservices, WASM, design patterns.

Data: PostgreSQL, MongoDB, Redis, OpenSearch, Kibana.

Cloud & Infra: AWS (EC2, S3, Secrets Manager), GCP, Terraform, Ansible, Docker, Kubernetes.

DevOps & Tools: Jenkins, GitHub Actions, CI/CD, Pytest, OpenTelemetry, Linux, Git.

Experience

Research Intern

Jul 2026 - Aug 2026

Corelight

Remote, USA

- Built (Python) an agentic platform and a custom 23-tool MCP server (Ghidra, radare2, capa, FLOSS, and androguard) for automated malware static analysis and reverse engineering across diverse file formats.
- Designed guardrails sandboxing the agent from raw samples, and iterated the pipeline to auto-generate YARA rules.
- Iterated agent prompting and tool orchestration across a ~38K-sample corpus (138 malware families, 6 file types) to raise detection coverage from **40-60% to 80-98%** and cut cost from **~\$10 to ~\$4** per family run.

Software Development Engineer (Intern to SDE)

Feb 2024 - Aug 2025

Zscaler

India

- Refactored legacy cloud modules with Composite and Factory design patterns, cutting technical debt by **~50%**.
- Built a Pytest platform running **900+ daily regression tests**, catching regressions within hours of a commit.
- Automated AWS/GCP provisioning with Terraform and Ansible, cutting environment setup from **18 hours to 3**.
- Automated the AWS sim-box lifecycle (EC2, S3, Secrets Manager) for **12x faster** setup, with OpenSearch/Kibana monitoring across 1,000+ metrics for real-time health checks and faster incident triage.
- Integrated Zscaler Private Access (ZPA) secure-access features adopted by 7+ internal teams, and reviewed 70+ pull requests for code quality, security, backward compatibility, and API-contract stability.

Projects

DepScan - Hybrid Dependency Upgrade Analysis Engine

Feb - Mar 2026

- Built a Go engine combining native AST call-graph extraction with LLM reasoning to judge dependency-upgrade safety across Go, Python, and Node.js codebases, flagging breaking API changes before merge.
- Shipped CI/CD integrations: GitHub Action annotations and Slack Block Kit webhooks.

HMCP - Secure MCP Middleware

Feb - May 2026

- Built quantum-safe MCP middleware in Rust with a gRPC transport layer, using a hybrid Kyber-768/X25519 KEM over a Noise XX handshake for forward-secret, tamper-evident channels between agents and tools.
- Added decentralized-identifier auth, rate limiting, and a 3-tier input sanitizer (homoglyph, mixed-script, control-char) to block prompt-injection, spoofing, replay, and man-in-the-middle attacks across the transport.

ZTBI - Zero-Trust Browser Interface

Nov - Dec 2025

- Built a Chrome extension running DistilBERT on-device (ONNX) to detect prompt injection and PII leaks in real time.
- Engineered a hybrid scoring engine hardened against tag-splitting and asynchronous exfiltration.

Additional

- **Publication:** "PSO-Driven Test Pattern Generation for Hardware Trojan Detection," IEEE 2024. DOI.
- 7 vulnerabilities discovered and disclosed to vendors; merged 2 PRs into IBM's qiskit-machine-learning.
- Competitive programming: peak CodeChef rating 1893; 500+ algorithmic problems solved.
- **Work Authorization:** F-1 OPT (STEM) - up to 36 months, no immediate sponsorship.